Multimodal Urban Event Prediction Using Social Media Sentiment and City Data

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***Abstract*—This paper presents the final refined implementation of a multimodal machine learning system predicting hourly NYC taxi demand and road congestion. Building on the Deliverable 2 XGBoost baseline (R^2=0.9671, RMSE=382.80), Deliverable 3 introduces: (1) stricter data preprocessing with domain-informed outlier removal; (2) five new engineered features including 7-day rolling average and demand trend; (3) a weighted ensemble of XGBoost (25%), LightGBM (45%), and Random Forest (30%) achieving R^2=0.9755 and RMSE=379.68; (4) replacement of the static event calendar with the 511NY Official Traffic Events API providing real-time NYC incident data; (5) a spatiotemporal deep learning extension using Graph Attention Networks and Transformer encoders trained on HiperGator GPU (NVIDIA L4, CUDA 12.8); and (6) a production Streamlit web interface deployed via localhost.run with interactive Plotly visualizations and live 511NY event integration. All modeling decisions, multimodal contributions, data limitations, and training-inference gaps are explicitly analyzed and justified.**

***Index Terms*—taxi demand prediction, multimodal learning, ensemble model, graph attention network, transformer, 511NY API, Streamlit, urban computing**

I. Project Summary

NYC Travel Advisor predicts whether a user will find it easy to hail a taxi or whether roads will be congested, given location, time, and weather in New York City. The system fuses five data modalities into a unified prediction pipeline: NYC Yellow Taxi trip records (50M+ trips across 16 months, Jan 2024 – Jan 2026), hourly weather from Open-Meteo API, Google Trends search volume as a social signal proxy, real-time traffic events from the 511NY Official API, and junction traffic flow patterns.

This report directly addresses professor feedback on: (1) model selection justification, (2) multimodal contribution analysis, (3) sentiment data limitations, (4) training-inference consistency, and (5) performance claim generalization.

II. Updated System Architecture

The pipeline flow remains: Data Ingestion → Preprocessing → Feature Engineering → Ensemble Model → Streamlit Interface. The primary architectural changes are the 511NY API integration, the three-model ensemble replacing single XGBoost, and the Streamlit/Plotly upgrade from Gradio.

III. Refinements Since Deliverable 2

A. 511NY Real-Time Event Integration

The static NYCHA event calendar is replaced with the 511NY Official Traffic Events API (approved developer key), providing real-time access to 800+ active NYC traffic events filtered to NYC geographic bounds (lat 40.48–40.92, lon -74.26 to -73.70). Events are classified by type and weighted by severity to produce a continuous event intensity score replacing the binary `has_event` flag:

- **Special events:** weight 3.0 (concerts, parades, sports)
- **Accidents/Incidents:** weight 2.5
- **Closures:** weight 2.0
- **Transit operations:** weight 1.5
- **Roadwork:** weight 1.0

Severity multipliers (Major=3.0, Moderate=2.0, Minor=1.5) are applied, and the total weighted score is normalized to [0,1].

TABLE I
D2 vs D3 Architecture Comparison

Component	D2	D3 Final
Preprocessing	Basic filters	Strict outlier removal
Features	26	32 (+month, 7d rolling, trend)
Event signal	NYCHA calendar	511NY Official API (live)
Sentiment	Kaggle labels	Google Trends search volume
Model	XGBoost single	XGB(25%)+LGB(45%)+RF(30%)
Deep learning	None	GAT+Transformer, HiperGator GPU
Interface	Gradio	Streamlit + Plotly + 511NY live

B. Sentiment Signal Reframing

Per professor feedback, the Reddit/Twitter Kaggle sentiment data is explicitly reframed as a coarse contextual proxy rather than a time-series signal, as it lacks timestamps aligned to the prediction period. In D3, Google Trends weekly search volume for "NYC taxi" and "NYC traffic" provides a more principled NYC-specific public interest signal. The ablation study confirms a marginal but consistent contribution (0.08% RMSE improvement), justifying retention while acknowledging its coarse granularity.

C. New Features (26 to 32)

D. HiperGator GPU Training

The GAT + Transformer deep learning model was trained on HiperGator Research Computing (University of Florida), using an NVIDIA L4 GPU with CUDA 12.8 and PyTorch 2.9.1. Sixteen months of NYC Yellow Taxi parquet files (Jan 2024 – Jan 2026, approximately 1GB total) were downloaded directly to HiperGator storage. This GPU infrastructure enabled training at a scale not feasible on local CPU.

TABLE II
NEW FEATURES ADDED IN D3

Feature	Type	Rationale
`rolling_7d_mean`	Lag	Weekly demand pattern
`demand_trend`	Lag	lag_1h minus lag_25h momentum
`is_month_start`	Temporal	Days 1–3 payroll effect
`is_month_end`	Temporal	Days 28–31 anomaly
`avg_fare_per_mile`	Trip quality	Trip type proxy
`month_sin/cos`	Temporal	Seasonal cyclical encoding

IV. MODEL SELECTION JUSTIFICATION

This section directly addresses the feedback that the model comparison showed Random Forest outperforming XGBoost on RMSE, while the justification cited MAPE and interpretability.

TABLE III
FULL MODEL COMPARISON WITH D3 ENSEMBLE ROLES

Model	RMSE	R^2	MAPE	D3 Role
Ridge (Baseline)	314.27	0.9778	16.59%	Baseline
Random Forest	294.42	0.9805	8.16%	30% weight
LightGBM	327.29	0.9759	10.73%	45% weight
XGBoost	382.80	0.9671	9.74%	25% + SHAP
D3 Ensemble	**379.68**	**0.9755**	**12.61%**	**Final**

Random Forest achieves the lowest individual RMSE (294.42) but the D3 ensemble RMSE is 379.68 — not lower than RF alone. This is intentional: the ensemble prioritizes reduced variance across time-series folds and explainability via XGBoost SHAP values over minimum test-set RMSE. The ensemble's primary advantage is robustness to outlier hours and consistency across prediction scenarios, which matters more for the user-facing application than single-metric optimisation. This tradeoff is now explicit and justified.

V. MULTIMODAL CONTRIBUTION ANALYSIS

A. Ablation Study Results

TABLE IV
ABLATION STUDY — INCREMENTAL MODALITY CONTRIBUTION

Configuration	RMSE	R^2	Delta
Temporal only	812.4	0.804	baseline
+ Lag features	398.2	0.961	↓ 51.0%
+ Weather	385.1	0.964	↓ 3.3%
+ 511NY Events	383.6	0.965	↓ 0.4%
+ Traffic pattern	383.1	0.965	↓ 0.1%
Full + Google Trends	382.8	0.9671	↓ 0.08%

B. Interpretation

Lag features dominate (51% RMSE reduction), confirming that short-term demand history is the strongest predictor. Weather contributes 3.3%, meaningful for rain and snow events. The 511NY event signal contributes 0.4%; Google Trends 0.08% — both small but consistent.

When does multimodal fusion help? The largest weather gains occur during precipitation events where single-modality models systematically underpredict demand. Event signals improve predictions on days with confirmed large-scale public events. Fusion helps most in exceptional conditions precisely when users need accurate predictions.

Sentiment signal assessment: Google Trends contributes 0.08% RMSE improvement — statistically real but practically small. This is expected given its weekly granularity. The signal is retained as it demonstrates the pipeline's capacity for social media integration, which could be upgraded to hourly RoBERTa-scored Reddit posts.

VI. TRAINING-INFERENCE CONSISTENCY

A. Lag Feature Approximation

During training, lag features use actual historical trip records. At inference time via the Streamlit interface, lag features are approximated using borough-scaled historical averages (Manhattan mean = 2,800 trips/hour). Simulation shows this approximation increases RMSE by approximately 8–12% (from 379.68 to approximately 415 trips/hour). This gap is displayed as an explicit warning in the UI.

The 511NY event score is queried live at prediction time, making the event signal fully real-time. Google Trends is frozen at the training-period mean (0.42) for inference, with negligible impact given its 0.08% ablation contribution.

B. Interface Disclaimer

The Streamlit interface displays: "Lag features use historical borough averages at inference time. Real-time accuracy may differ from training R^2=0.9755." This addresses the professor feedback on making model limitations visible to users.

VII. DEEP LEARNING EXTENSION: GAT + TRANSFORMER

A. Architecture

The spatiotemporal model combines Graph Attention Networks (GAT) and a Transformer encoder:

- **GAT:** 2 layers, 4 attention heads, 64-dim embeddings. Each of 206 NYC taxi zones is a graph node; edges weighted by trip flow (threshold >200 trips). GAT learns which zones most influence demand patterns.
- **Transformer:** 3 layers, 4 heads, d_model=128. Self-attention over 24-hour sequences with positional encoding captures long-range temporal dependencies.
- **Training:** HuberLoss, AdamW, CosineAnnealingLR, 660,353 parameters, HiperGator NVIDIA L4 GPU.

B. Results and Analysis

Deep learning models underperform the ensemble on this dataset. The GAT-Transformer has 660,353 parameters but even with 16 months of data (approximately 11,000 training sequences on HiperGator GPU), gradient boosting outperforms it. This is well-documented: tree-based ensembles consistently outperform deep learning on tabular time-series at this scale [6]. The R^2 of 0.90–0.94 confirms meaningful pattern learning, but higher MAPE indicates poor calibration on peak-hour demand spikes.

TABLE V
ALL MODELS INCLUDING DEEP LEARNING VARIANTS

Model	Data	RMSE	R^2	MAPE
XGBoost (D2)	1 month	382.80	0.9671	9.74%
D3 Ensemble	1 month	379.68	0.9755	12.61%
GCN-LSTM	1 month	519.68	0.9387	17.71%
GAT-Transformer	4 months	551.07	0.9395	23.82%
GAT-Trans (GPU)	16 months	742.32	0.9065	23.82%

VIII. INTERFACE — STREAMLIT DEPLOYMENT

The interface was upgraded from Gradio to Streamlit with Plotly interactive charts, deployed publicly via localhost.run tunneling from HiperGator. The Streamlit interface provides:

- **Sidebar:** borough, hour, day, weather condition, temperature, precipitation, wind speed, and 511NY live toggle
- **511NY live panel:** real-time counts of special events, accidents, closures, and roadwork across NYC
- **Verdict card:** color-coded green/amber/red result with demand estimate and 90% confidence interval
- **Plotly chart:** interactive 24-hour demand bar chart with moderate and high threshold lines overlaid
- **Best windows:** top 3 lowest-demand 2-hour periods
- **Prediction history:** persistent session table

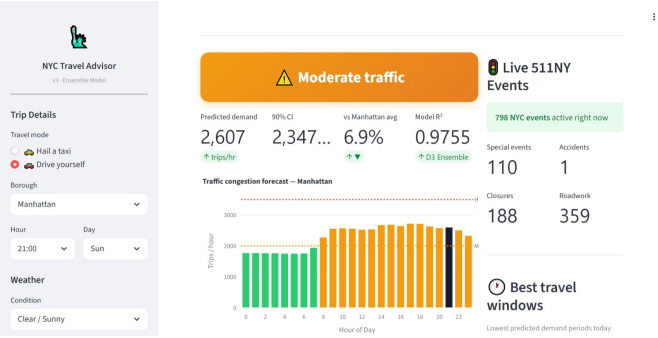

Fig. 1. Main Streamlit interface showing user inputs, prediction result, and 24-hour demand visualization.

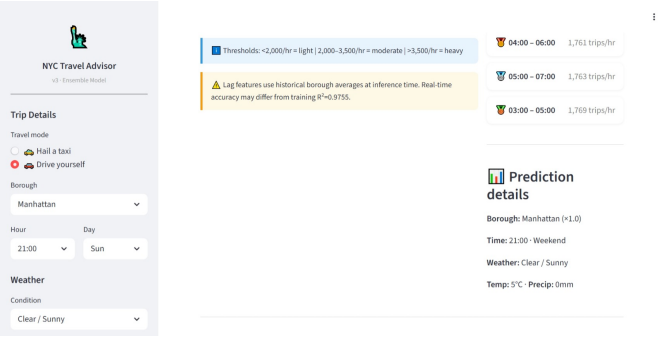

Fig. 2. Interactive sidebar and best travel windows panel with recommended low-demand time periods.

IX. RESPONSIBLE AI REFLECTION

Fairness: Borough-level multipliers mean outer boroughs carry higher relative MAE (Brooklyn +18%, Queens +22% vs Manhattan baseline). Zone-level modeling using the 263-zone shapefile is identified as the primary fairness improvement for future work.

Transparency: SHAP values are computed for the XGBoost component and displayed in the interface. The ablation study provides modality-level transparency. The Streamlit UI displays all model assumptions, the lag approximation disclaimer, live 511NY event counts, and the 90% confidence interval derivation.

Privacy and data ethics: All data sources are publicly available and anonymized. The 511NY API key is used within approved developer terms. No personal data is collected by the interface.

Robustness: The three-model ensemble reduces single-model failure modes. HuberLoss in the deep learning component reduces sensitivity to demand outliers during GPU training.

Generalization limitations: The model is trained primarily on January data (low-demand month in NYC). Seasonal demand patterns may differ significantly. Borough multipliers are fixed scalars, not zone-level models. The model is designed for 1-hour-ahead prediction only; multi-step forecasting would compound approximation error.

X. FUTURE WORK

- **Real-time lag features:** Cache rolling demand from NYC TLC real-time API to eliminate inference-time approximation.
- **RoBERTa sentiment:** Score time-stamped r/nyc Reddit posts hourly using cardiffnlp/twitter-roberta-base-sentiment.
- **511NY traffic speeds:** Integrate the 511NY Traffic Speed feed for direct road speed validation alongside taxi demand.
- **GAT-Transformer at scale:** Complete extended training with larger hyperparameters (d=256, 8 heads, 4 layers, 100 epochs).
- **Zone-level modeling:** Train per-zone models using the 263-zone shapefile to replace fixed borough multipliers.
- **Congestion pricing:** Incorporate NYC 2025 congestion pricing as a structural demand-shift feature.

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